

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Device Risk Analysis | Case Study No. 86

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1. System Purpose & Architecture

This document specifies the software architecture, data schemas, functional requirements, and operating constraints for the Device Risk Analysis software. The solution implements a complete supervised machine learning workflow encapsulated within a modern Streamlit user interface.

2. Telemetry Schema & Data Dictionary

Field Name	Type	Domain Range	Null Imputation
Device_Type	Categorical	6 classes (Laptop, Server, etc.)	Mode
Operating_System	Categorical	11 classes (Win11, Linux, etc.)	Mode
Device_Age_Months	Integer	1 to 72 months	Median
Firmware_Age_Months	Integer	0 to 36 months	Median
Failed_Login_Attempts	Integer	0 to 50 attempts	Median
Open_Ports_Count	Integer	1 to 25 ports	Median
Vulnerability_Count	Integer	0 to 20 CVEs	Median
Encryption_Enabled	Categorical	Yes, No	Mode
Antivirus_Status	Categorical	Active, Outdated, Disabled	Mode
Risk_Level (Target)	Categorical	Low Risk, Medium Risk, High Risk	Supervised Target

3. Functional Requirements Summary

ID	Module	Description
FR-01	Ingestion	Load raw CSV telemetry validating column headers
FR-02	Cleaning	Automated duplicate deletion and missing value imputation
FR-03	Pipeline	ColumnTransformer with StandardScaler & OneHotEncoder
FR-04	Benchmark	Train and compare 6 classification models (F1-weighted)
FR-05	Tuning	5-fold Stratified GridSearchCV for optimal regularization
FR-06	Web App	6-page Streamlit application for on-demand predictions

4. Software & Runtime Constraints

Python Version: Python 3.12 within dedicated virtual environment.

Fault Tolerance: Defensive loading handles missing model files gracefully without application termination.